Tiny-YOLOSAM

“Fast, reliable, and accessible image segmentation”

Group 1: Katherine Zhang, Abby Schultz, Mason Wu, Max Han, and Kenneth Xu

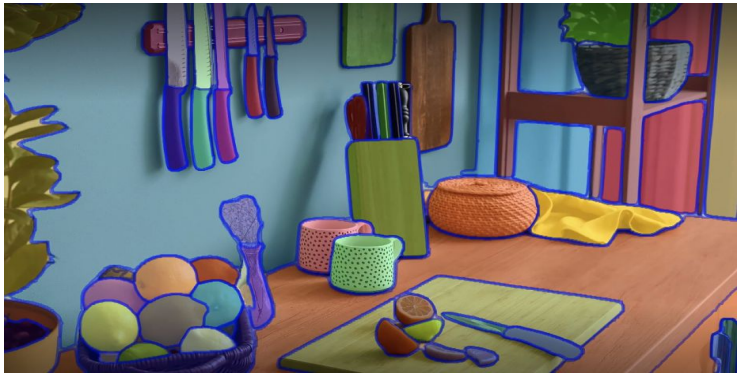
Background: Introduction to AI Models and Image Segmentation

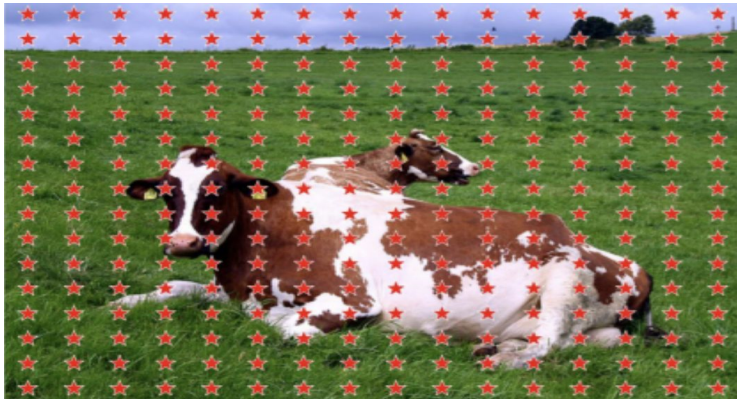
State of the Art:

- SAM: strong universal segmentation
- YOLO: fast semantic detection
- TinySAM: lighter SAM variant

Issues:

- SAM extremely slow inference
- YOLO misses unlabeled objects
- TinySAM still runs on costly prompts





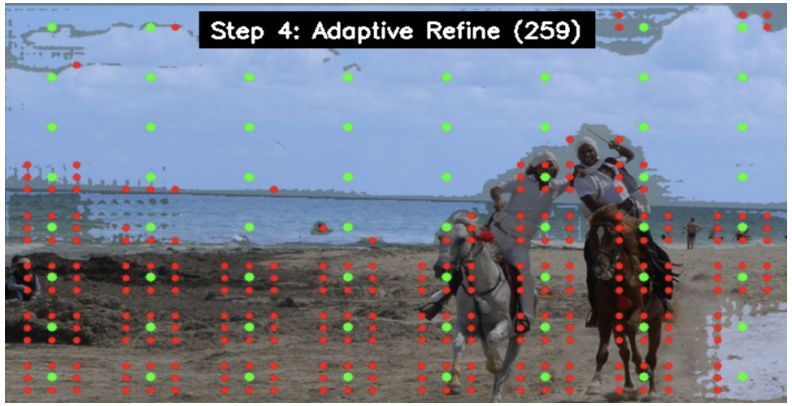
Tiny-SAM

Methodology Crash Course:

- Distills SAM into student
- Quantizes encoder + decoder
- Hierarchical point prompting
- Uses COCO training data

Takeaways/Features:

- Much faster than SAM
- Segmentation quality still good
- Still needs dense prompts
- Works zero-shot on images





TinySAM Replication

Reproduced Average Precision (AP):

- Evaluation done on full COCO 2017 Dataset

Non-Quantized Model

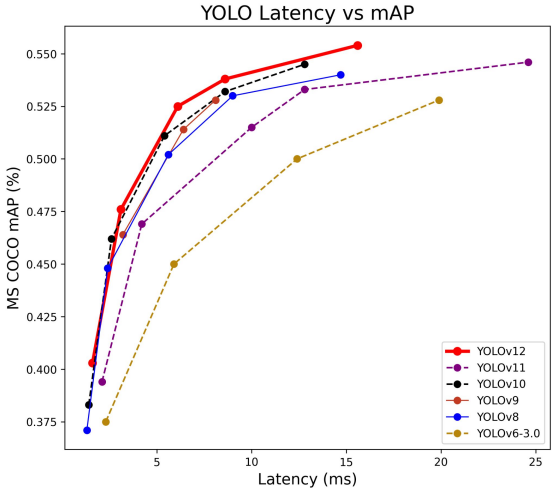
- 42.27% (Paper claims 42.3%)

Quantized Model

- 41.37% (Paper claims 41.4%)



YOLO-V12

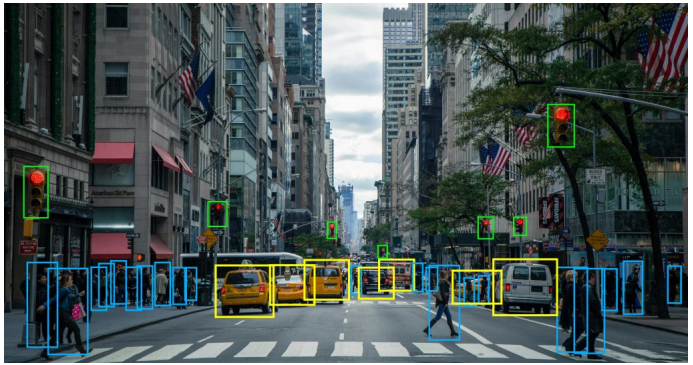


Methodology:

- Transformer instead of CNN in YOLO 11
- Area Attention to reduce computational cost in self-attention mechanism
- Flash Attention to conquer the memory access issue of attention

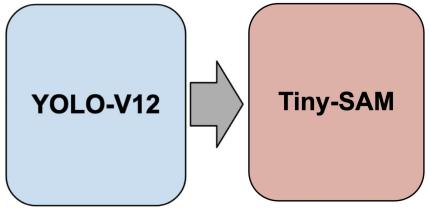
Takeaways/Features:

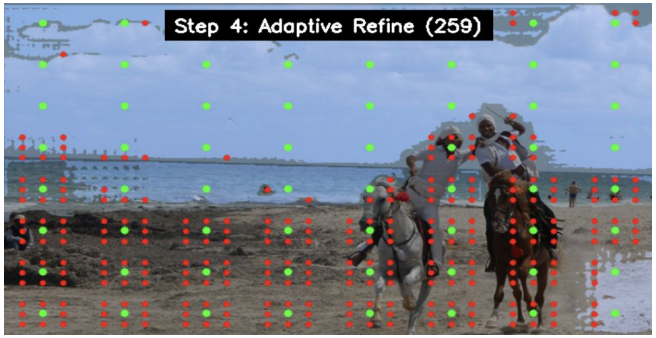
- Instant, Fast, Bounding Boxes
- We use YOLO-V12s Turbo for their bounding boxes



Extension Motivation

- MobileSAMv2 shows that limiting SAM to object-discovered prompts cuts computation by 16× and improves mask proposals.
- We follow the same idea: YOLO detections → prompts for TinySAM, enabling efficient segment-everything performance.





source: https://arxiv.org/abs/2312.09579 (MobileSAMv2: Faster Segment Anything to Everything)

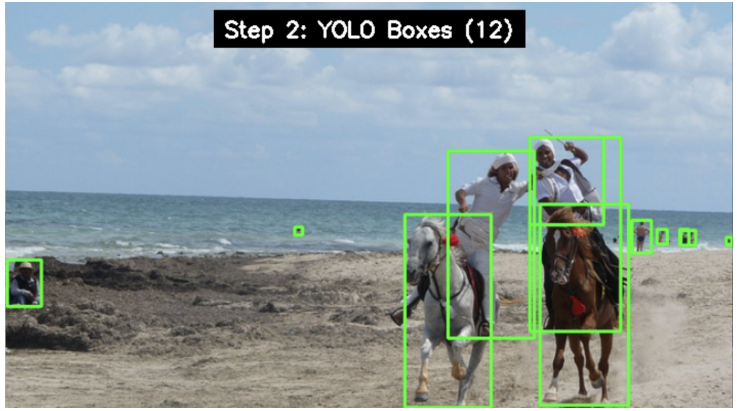
Naive Extension: Tiny-YOLOSAM

Inspiration:

- Everything mode: slow, impractical
- YOLO: fast but coarse boxes
- Box prompts yield accurate pinpoint masks

Implementation:

- YOLO finds prominent objects
- Draw box around each object
- TinySAM traces exact object shape





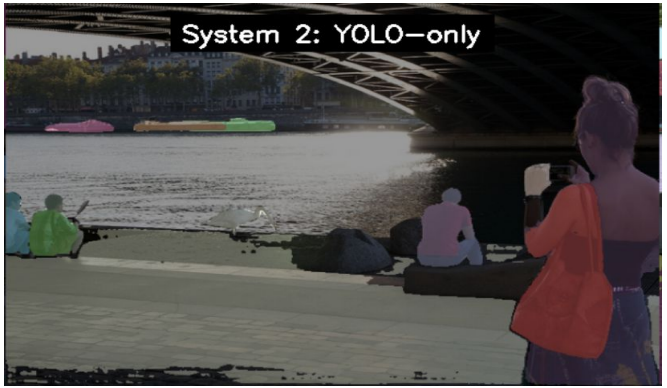
Mature Extension: Hybrid Tiny-YOLOSAM

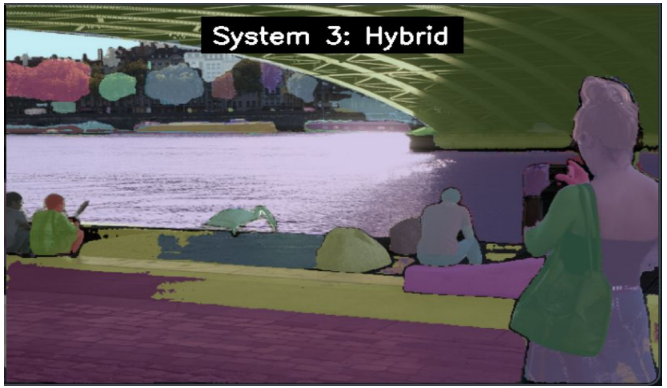
Inspiration:

- YOLO-only misses background objects
- Everything mode covers all regions
- YOLO fast but incomplete coverage
- Sparse point map fill missed areas

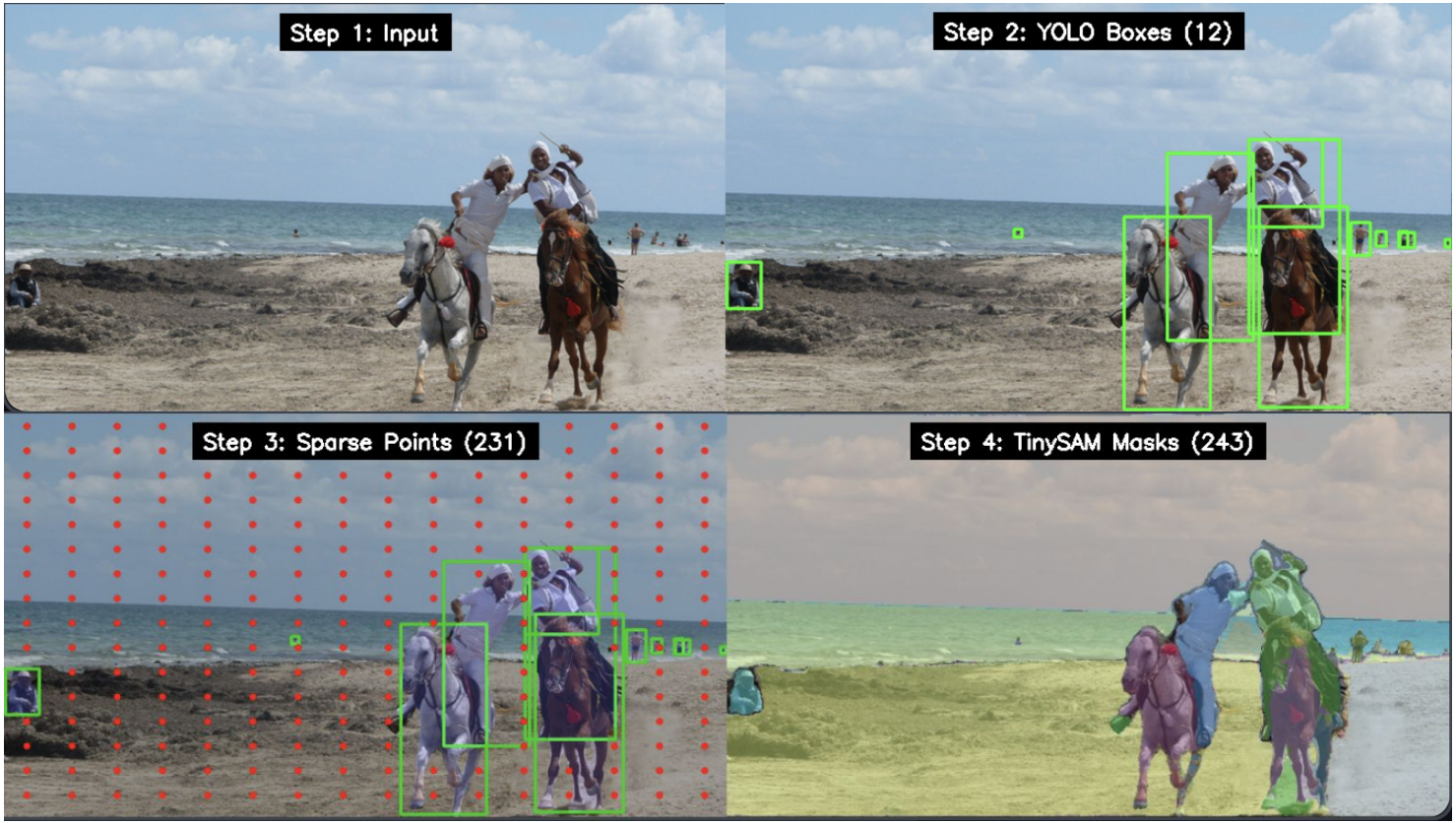
Implementation:

- YOLO identifies prominent objects
- Build coverage mask from YOLO
- Sample sparse grid excluding coverage
- TinySAM segments uncovered point prompts





Full Hybrid Pipeline Visualization



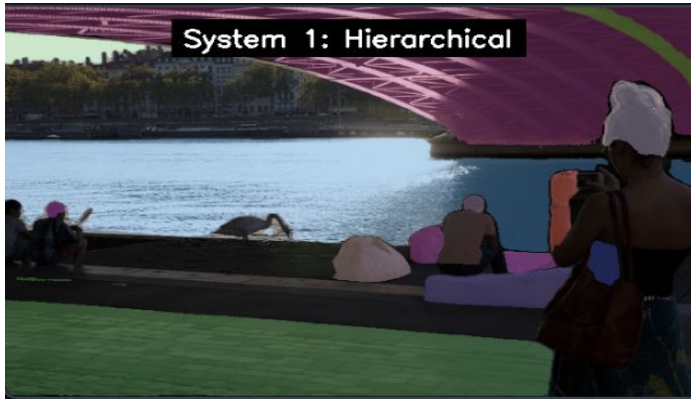
Results and Evaluation

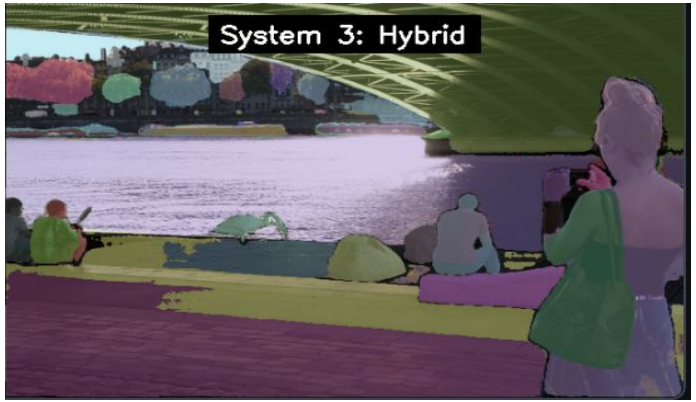
Metric Comparisons:

- Hierarchical Point Sampling baseline: 31-44s per Image (adaptive overhead)
- Hybrid speedup: 6-8s, 3-8 times faster
- Quality preserved: AP 34%
- Coverage: Average Recall 76% vs 18% hierarchical baseline

Takeaways:

- Avoids expensive adaptive algorithms
- YOLO handles foreground, sparse fills





Why Better?

Algorithmic Advantages

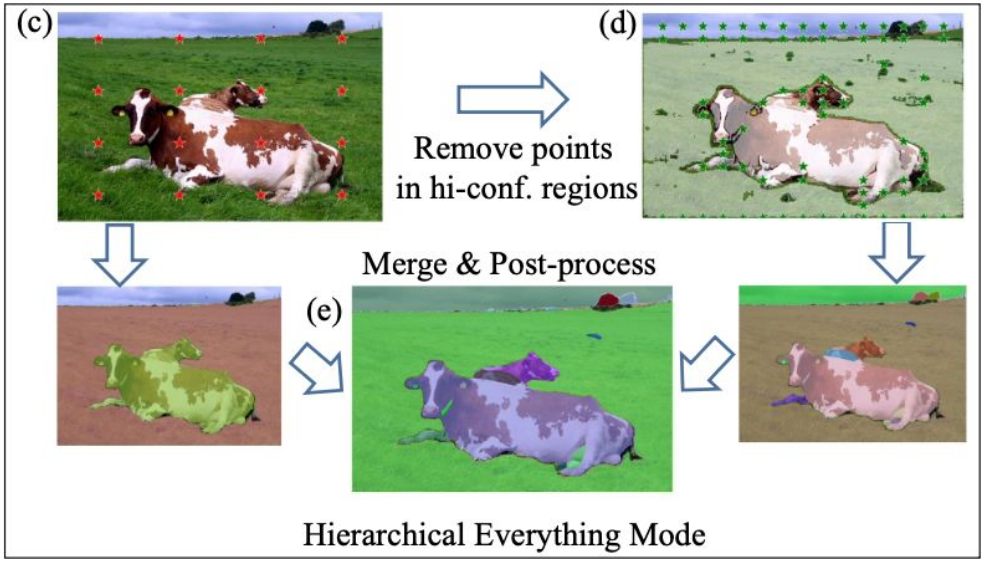
- Single-pass efficient detection
- Semantics-guided “segment everything”





Engineering Advantages

- Modular, Interpretable Design



- Research and Reproducibility-Friendly

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Algorithmic Advantages

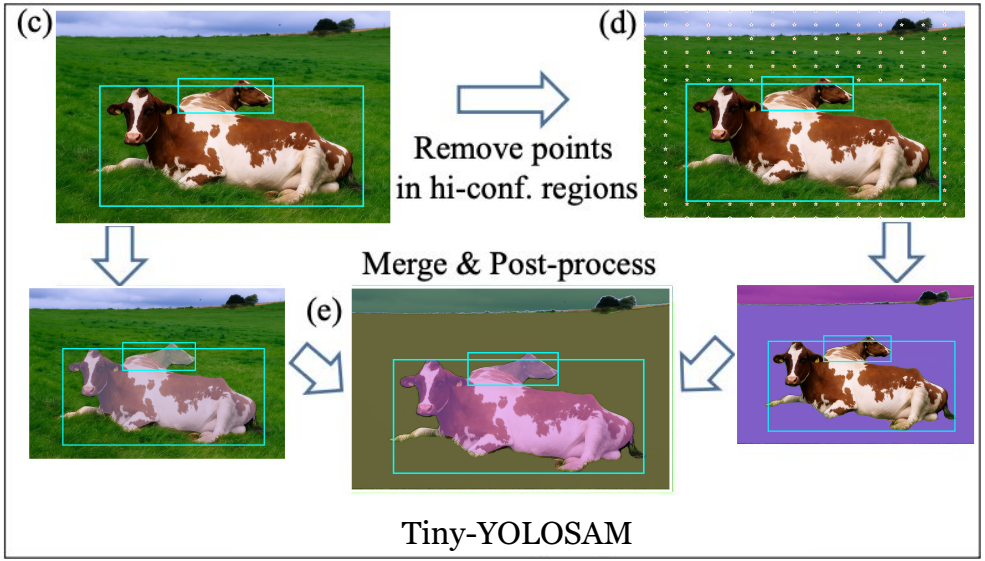
- Single-pass efficient detection
- Semantics-guided “segment everything”





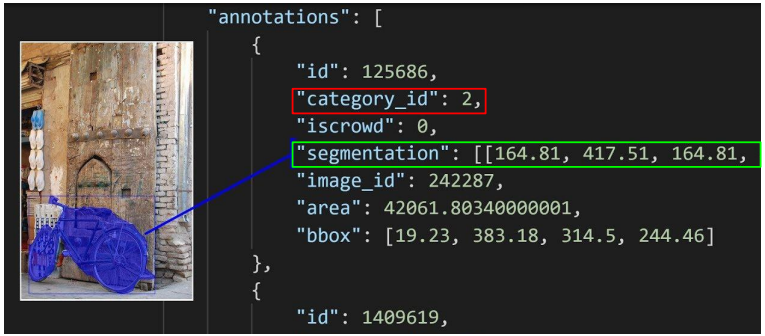
Engineering Advantages

- Modular, Interpretable Design



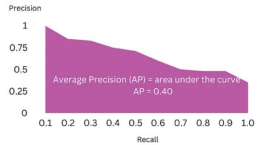
- Research and Reproducibility-Friendly

Evaluation (500 COCO Images)

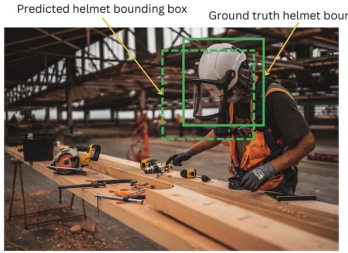


For **foreground objects** with yolo bounding box, we have category_id to compare, making COCO metrics feasible (**AP**).

For **background masks** generated by sparse points, we choose to use **Class-Agnostic metrics, such as AR and mIoU**.

Metrics: AP 

, AR, IoU



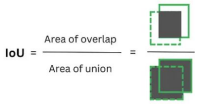
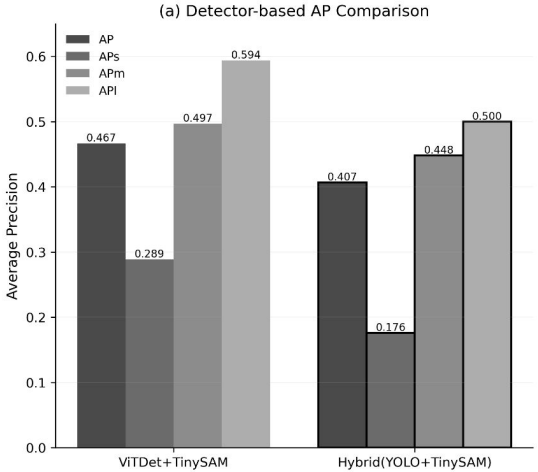


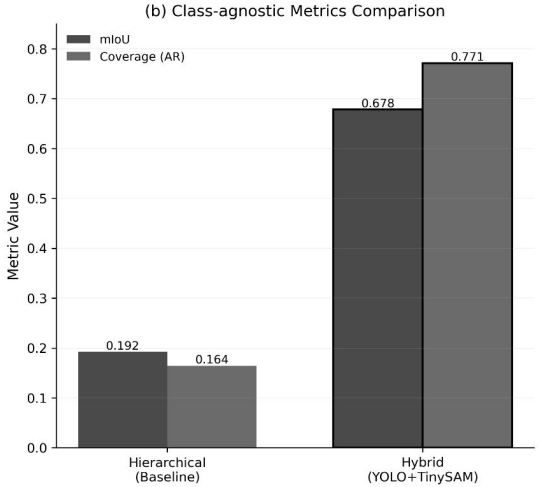
Table X: Detector-based evaluation (COCO val2017 subset, 500 images)

```
================================================================================
DETECTOR-BASED COMPARISON (with category labels)
================================================================================
System                          AP       APs      APm      APl      Time/img   Calls/img
--------------------------------------------------------------------------------
1. ViTDet→TinySAM (baseline)    0.467    0.289    0.497    0.594    1.63       17.6
2. YOLO→TinySAM                 0.407    0.176    0.448    0.500    0.38       8.1
3. Hybrid (YOLO boxes only)     0.407    0.176    0.448    0.500    7.19       174.2
================================================================================
Note: Hybrid AP computed from YOLO box masks only (sparse points are class-agnostic)
```

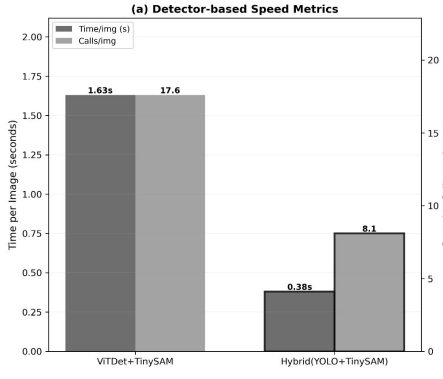
Table Y: Class-agnostic comparison (same subset, coverage-focused)

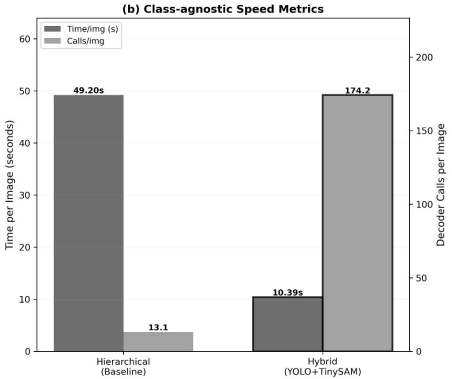
```
================================================================================
CLASS-AGNOSTIC COMPARISON (Hierarchical vs Hybrid)
================================================================================
System                          AR       mIoU     Coverage   Time/img   Calls/img
--------------------------------------------------------------------------------
Hierarchical (baseline)         0.164    0.192    0.164      289.78     13.1
Hybrid (YOLO+sparse)            0.771    0.678    0.771      8.05       174.2
================================================================================
```





Note: Hybrid and YOLO→TinySAM show identical performance because they use the same YOLO box predictions for AP evaluation.





Limitation and Future work

- **Limitation:** On images with only large objects, our Hybrid model outputs a single whole-object mask, while original TinySAM produces many fine-grained parts, making them hard to compare.
- **Potential Fix:** Use CLIP to convert YOLO labels into prompts so TinySAM also predicts whole-object masks.
- **Future Work:** Address the current mAP drop through improving recall and category coverage—especially for low-frequency classes (etc. books and handbags).







Thank You!